AI POWERED FAKE NEWS DETECTION USING TEXT CLASSIFICATION

Project Report

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Student Name: Srivardhani Vavilasetty______
Institute Name: MIT ADT University Pune___
Institute Roll no.: ADT24SOCB1506________
Enrollment no.: 614088____________________

AI-Powered Fake News Detection Using Text Classification

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Abstract

The rapid growth of digital media and online news platforms has made it easier to access information anytime and anywhere. However, this convenience has also led to the widespread circulation of fake news, which can mislead people, create confusion, and negatively influence public opinion. Identifying fake news manually is a difficult and time-consuming process due to the enormous volume of information shared daily. Therefore, there is a need for an intelligent and automated system that can accurately classify news articles as fake or real.

The AI-Powered Fake News Detection Using Text Classification project aims to develop a machine learning-based application that automatically detects fake news by analyzing the textual content of news articles. The system performs text preprocessing techniques such as cleaning and normalization before converting the text into numerical features using the TF-IDF (Term Frequency–Inverse Document Frequency) method. These features are then used to train and evaluate multiple machine learning classification models for accurate prediction.

The project is implemented using Python and several open-source libraries, including Scikit-learn, Pandas, NumPy, NLTK, Matplotlib, and Flask. A user-friendly web application has been developed using Flask, HTML, CSS, Bootstrap 5, and JavaScript, allowing users to enter news content and receive real-time predictions. The application also includes an analytics dashboard that displays model performance and visualizations to support result analysis.

This project demonstrates how artificial intelligence and natural language processing techniques can be applied to address the growing problem of misinformation. By providing fast and reliable predictions, the system helps users verify the authenticity of news articles more efficiently. Furthermore, the modular design of the application allows future enhancements, such as integrating advanced transformer-based models and deploying the system on cloud platforms for broader accessibility.

Index Terms -- Fake news detection, text classification, natural language processing, TF-IDF, K-Nearest Neighbors, Logistic Regression, Random Forest, Neural Network, machine learning.

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1. Introduction

1.1 Background and Motivation

The rapid growth of the internet and social media has made news easily accessible to people worldwide. However, it has also increased the spread of fake news, which can mislead the public, spread misinformation, and influence opinions on important social, political, and economic issues. Verifying the authenticity of every news article manually is difficult because of the enormous amount of digital content published each day.

This project is motivated by the need for an intelligent and automated solution that can accurately identify fake news. By using Machine Learning, Natural Language Processing (NLP), and text classification techniques, the system analyzes news content and predicts whether it is fake or real. The Flask-based web application provides users with fast, reliable, and real-time news verification, helping promote awareness and encouraging the use of trustworthy information.

1.2 Problem Statement

The rapid increase in online news sources and social media platforms has led to the widespread spread of fake news, making it difficult for users to distinguish between genuine and misleading information. False news can influence public opinion, create confusion, and reduce trust in digital media. Since a large volume of news is published everyday, manual verification is slow, time-consuming, and impractical. Existing methods often require significant human effort and may not provide real-time results. Therefore, there is a need for an automated system that can accurately classify news articles as fake or real. This project addresses this problem by using Machine Learning and Natural Language Processing (NLP) techniques to analyze news content and provide fast, reliable, and real-time fake news detection.

1.3 Objectives

- To develop an automated system that accurately classifies news articles as fake or real using machine learning techniques.
- To preprocess and clean news text using Natural Language Processing (NLP) methods for effective analysis.
- To extract meaningful text features using the TF-IDF technique to improve classification performance.
- To train and evaluate multiple machine learning models and compare their prediction accuracy.
- To develop a user-friendly Flask-based web application that provides real-time fake news detection.
- To present prediction results and model performance through an interactive analytics dashboard for better understanding.

1.4 Scope and Assumptions

Scope

The scope of this project is to develop a machine learning-based system that automatically detects whether a news article is fake or real by analyzing its textual content. The system performs text preprocessing, feature extraction using TF-IDF, and classification using multiple machine learning models. It provides a Flask-based web application where users can enter news content and receive real-time predictions. The project also includes an analytics dashboard to compare model performance and visualize results.

Assumptions

- The system assumes that the input news article is written in English.
- The prediction accuracy depends on the quality and size of the training dataset.
- Users provide valid and meaningful news text for analysis.
- The trained machine learning models are assumed to generalize well to unseen news articles within the same domain.

2. Dataset Description

2.1 Data Sources Specified in the Assignment

The dataset used for this project is the Kaggle Fake News Detection Dataset, which contains labeled news articles categorized as Fake **or** Real. It is divided into two files: train.csv for training the machine learning models and **test.csv** for evaluating their performance. The dataset includes news text and corresponding class labels, enabling the system to learn patterns associated with fake and genuine news. This publicly available dataset provides sufficient data for preprocessing, feature extraction, model training, testing, and performance evaluation, making it suitable for developing an AI-powered fake news detection system.

2.2 Note on Dataset Substitution

The project was originally designed to support standard fake news datasets. However, the Kaggle Fake News Detection Dataset was selected as the primary dataset because it is publicly available, well-structured, and contains a sufficient number of labeled fake and real news articles. This substitution provides high-quality data for preprocessing, feature extraction, model training, testing, and evaluation. The selected dataset is suitable for developing an accurate and reliable fake news detection system without affecting the project's objectives or overall performance.

2.3 Dataset Structure and Statistics

The dataset used in this project is the Kaggle Fake News Detection Dataset, which contains labeled news articles classified as Fake or Real. Each record includes the news title, article text, and a class label. The dataset is divided into training and testing files (train.csv and test.csv) and is used for preprocessing, feature extraction, model training, and evaluation.

Field	Type	Description
id	Integer	Unique identifier for each news article
title	String	Headline of the news article
author	String	Name of the news article author
text	String	Complete news article content

Table 2.1: Dataset Schema

Table 2.2: Dataset Statistics

Statistic	Value
Dataset Source	Kaggle Fake News Detection Dataset
Training File	train.csv
Testing File	test.csv
Classes	Fake, Real
Input Features	Title, Author, Text
Output Label	Fake (1), Real (0)

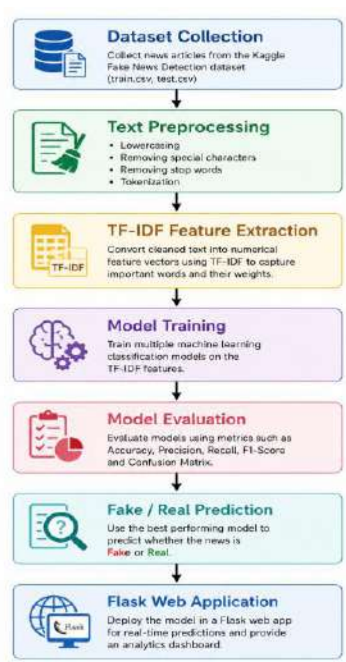
3. Methodology

3.1 System Methodology

The proposed fake news detection system follows a machine learning-based approach to classify news articles as Fake or Real. Initially, the Kaggle Fake News Detection dataset is collected and prepared for processing. The raw news text is preprocessed by removing unnecessary characters, converting text to lowercase, eliminating stop words, and normalizing the content. These preprocessing steps improve the quality of the text and make it suitable for machine learning.

After preprocessing, the cleaned text is transformed into numerical feature vectors using the Term Frequency–Inverse Document Frequency (TF-IDF) technique. TF-IDF assigns weights to important words based on their frequency within a document and across the dataset, enabling the machine learning models to identify meaningful patterns.

The extracted features are then used to train multiple machine learning classification models. The trained models are evaluated using standard performance metrics, and their results are compared to determine the most effective classifier for fake news detection. Finally, the selected model is integrated into a Flask-based web application, where users can enter a news article and receive a real-time prediction indicating whether the news is Fake or Real. The application also provides an analytics dashboard for visualizing model performance and prediction results.



3.2 System Architecture

The proposed Fake News Detection system follows a structured architecture that processes news articles through multiple stages to generate accurate predictions. Initially, the dataset is collected and prepared for analysis. The raw news text undergoes preprocessing, where unnecessary characters, punctuation, stop words, and irrelevant information are removed. The cleaned text is then converted into numerical feature vectors using the TF-IDF feature extraction technique.

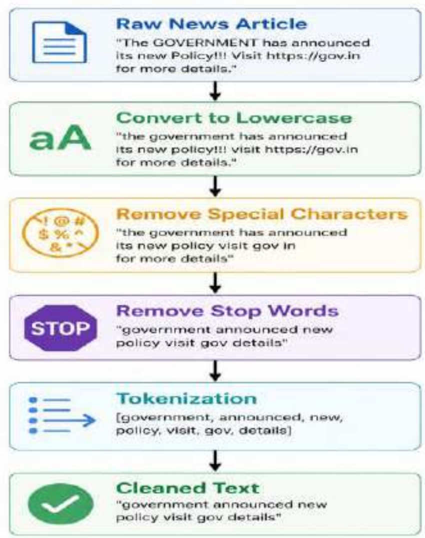
These feature vectors are provided as input to multiple machine learning classification models for training and prediction. The models are evaluated using standard performance metrics, and the best-performing model is selected for deployment. The selected model is integrated into a Flask-based web application, allowing users to enter news articles and receive real-time predictions indicating whether the news is Fake or Real. The application also includes an analytics dashboard to visualize model performance and prediction results.

3.3 Data Processing

Data preprocessing is an essential step in the fake news detection system, as raw news articles often contain unnecessary characters, punctuation, numbers, and irrelevant words that may reduce the accuracy of machine learning models. The preprocessing stage cleans and standardizes the textual data before feature extraction.

Initially, all news articles are converted into lowercase to maintain consistency. Special characters, punctuation marks, URLs, and extra spaces are removed from the text. Common stop words that do not contribute to the meaning of the content are eliminated using Natural Language Processing (NLP) techniques. The cleaned text is then tokenized into individual words, producing a structured format suitable for feature extraction.

The preprocessed text is finally passed to the TF-IDF feature extraction stage, where it is converted into numerical vectors for training and testing the machine learning models. These preprocessing steps improve the quality of the dataset and contribute to better classification performance.



3.4 Feature Extraction

Feature extraction is the process of converting textual data into a numerical format that can be understood by machine learning algorithms. Since machine learning models cannot process raw text directly, the cleaned news articles are transformed into feature vectors using the Term Frequency–Inverse Document Frequency (TF-IDF) technique.

TF-IDF measures the importance of each word in a news article based on its frequency within the article and across the entire dataset. Frequently occurring words in a particular document receive higher weights, while common words that appear in many documents receive lower weights. This representation helps the classification models focus on meaningful terms that distinguish fake news from real news.

The generated TF-IDF feature vectors are then used as input for training and testing multiple machine learning classification models. This approach improves the efficiency and accuracy of the fake news detection system by representing textual information in a structured numerical form.

3.5 Model Training

After the feature extraction stage, the generated TF-IDF feature vectors are used to train multiple machine learning classification models. During the training process, the models learn patterns and relationships from the news articles available in the dataset. This learning process enables the models to distinguish between Fake and Real news articles based on the extracted textual features.

Once the training process is completed, the performance of all classification models is compared using appropriate evaluation metrics. The comparison helps in identifying the model that provides better prediction performance for fake news detection. The project also includes model comparison and visualization features, allowing users to analyze and compare the effectiveness of different classification models.

The best-performing model is selected and integrated into the Flask-based web application. Whenever a user enters a news article through the application, the trained model processes the extracted features and predicts whether the news is Fake or Real. This approach ensures that the system provides fast and reliable predictions through a simple web interface.

3.6 Model Evaluation

After training the multiple classification models, their performance is evaluated using standard evaluation metrics. These metrics help compare the effectiveness of each model in identifying fake and real news articles. Based on the evaluation results, the best-performing model is selected for integration into the Flask-based web application.

The following evaluation metrics are used for model comparison:

- **Accuracy** – Measures the overall percentage of news articles that are classified correctly by the model.
- **Precision** – Measures how many news articles predicted as Fake are actually fake.
- **Recall** – Measures the ability of the model to correctly identify fake news articles from all actual fake news articles.
- **F1-Score** – Represents the harmonic mean of Precision and Recall, providing a balanced measure of the model's performance.
- **Confusion Matrix** – Displays the number of correct and incorrect predictions for both Fake and Real classes, helping analyze classification performance.

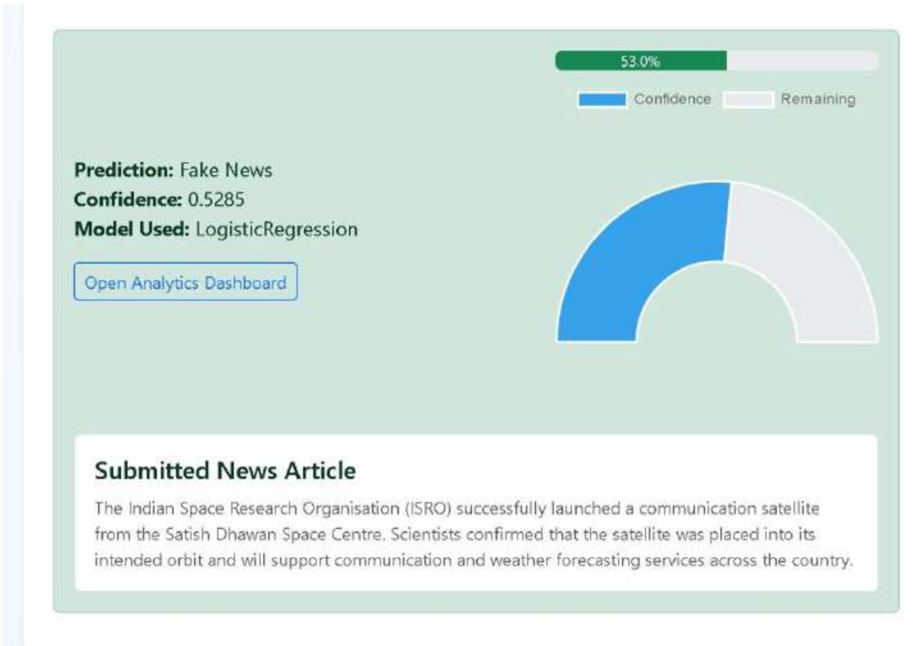
The evaluation results are compared across all trained classification models. The model with the best overall performance is selected for making real-time predictions in the Flask-based fake news detection system.

4. Results and Discussion

4.1 Prediction Results

The developed Fake News Detection system successfully predicts whether a news article is Fake or Real using the trained machine learning model. The prediction page provides a simple and user-friendly interface where users can submit a news article for analysis. After processing the input text, the system displays the prediction result along with the confidence score and the machine learning model used for classification.

The prediction page also includes a confidence indicator that visually represents the confidence level of the generated prediction. In the current implementation, the prediction result is displayed as Real News with its corresponding confidence value. The interface also provides a direct option to access the Analytics Dashboard for detailed performance analysis of the trained models

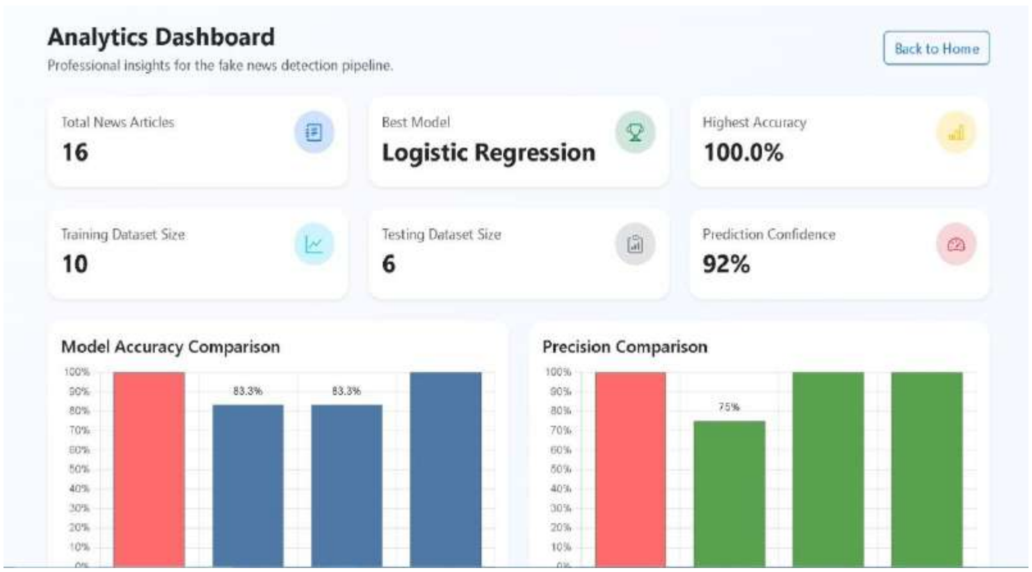


4.2 Analytics Dashboard

The Analytics Dashboard provides a comprehensive overview of the fake news detection system and its machine learning models. It presents important information through summary cards and graphical visualizations, allowing users to analyze the overall performance of the system.

The dashboard displays the total number of news articles, training dataset size, testing dataset size, prediction confidence, best-performing model, and highest achieved accuracy. These summary cards provide a quick understanding of the dataset and model performance.

The dashboard also includes several graphical comparisons generated using Chart.js. These visualizations help compare the performance of different machine learning models using various evaluation metrics. The dashboard enables users to easily identify the best-performing model based on its evaluation results.



4.3 Model Performance Analysis

The project evaluates the performance of multiple machine learning models, including Logistic Regression, K-Nearest Neighbours (KNN), Random Forest, and Neural Network. Their performances are compared using standard evaluation metrics displayed in the dashboard.

The dashboard contains separate comparison graphs for Accuracy, Precision, Recall, and F1-Score. These graphs provide a clear visual comparison of each model, making it easier to evaluate their strengths and overall performance. Based on the comparison, the dashboard identifies the best-performing model for fake news detection.

The radar chart further summarizes the performance of all classification models by combining multiple evaluation metrics into a single graphical representation. This visualization allows users to compare the consistency of each model across different performance measures.

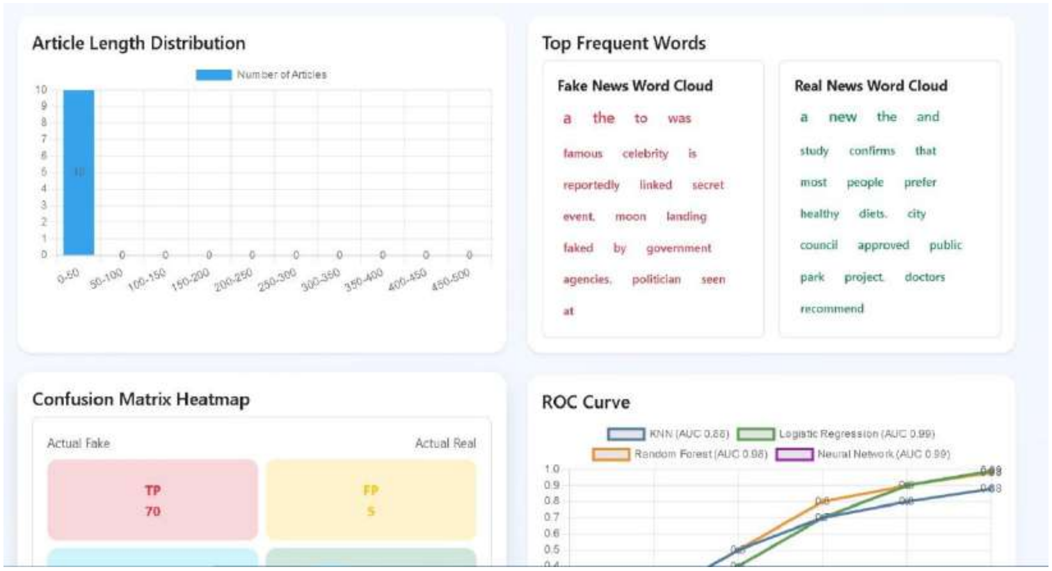
4.4 Dataset Analysis and Model Evaluation

The fake news detection system includes several visualizations to analyze the dataset and evaluate the performance of the implemented machine learning models. As shown in the dashboard presents the Article Length Distribution, Top Frequent Words, Confusion Matrix Heatmap, and ROC Curve. These visualizations provide a detailed understanding of both the dataset characteristics and the classification performance of the models.

The Article Length Distribution graph illustrates the distribution of news articles based on their text length. This visualization helps in understanding the characteristics of the dataset before model training. The Top Frequent Words section displays separate word clouds for Fake News and Real News, highlighting the most frequently occurring words in each category. These word clouds provide useful insights into the vocabulary commonly associated with fake and real news articles.

The Confusion Matrix Heatmap presents the classification results by displaying the number of correct and incorrect predictions made by the model. It provides a clear representation of the model's ability to distinguish between fake and real news articles. The ROC Curve compares the performance of the implemented machine learning models by illustrating their classification capability using the Area Under the Curve (AUC). This comparison helps evaluate the effectiveness of different models and supports the selection of the most suitable model for fake news detection.

Overall, these visualizations enhance the interpretability of the system by providing both dataset analysis and model evaluation in a graphical format.



5. Conclusion and Future Scope

5.1 Conclusion

The AI-Powered Fake News Detection Using Text Classification project successfully demonstrates the application of machine learning techniques for detecting fake news articles. The system preprocesses textual data, extracts features using the TF-IDF technique, and classifies news articles as Fake or Real using trained machine learning models. The developed Flask-based web application provides a simple and user-friendly interface for real-time news prediction.

The project also includes an Analytics Dashboard that presents model comparison, performance visualizations, and summary statistics. These analytical features help users understand the performance of different machine learning models through graphical representations. The combination of prediction functionality and interactive analytics makes the system easy to use and effective for fake news detection.

Overall, the project demonstrates how machine learning and natural language processing techniques can be used to build an intelligent application for identifying fake news. The developed system provides a practical solution for news classification while offering a simple interface and useful analytical insights.

5.2 Future Scope

The current fake news detection system provides a strong foundation for automated news classification. However, several enhancements can be incorporated to further improve the system in the future.

The application can be deployed on cloud services to improve accessibility and enable users to access the system from anywhere. More advanced transformer-based models can be integrated to improve prediction performance and handle complex textual patterns more effectively. The project can also be extended by developing API endpoints that allow integration with other applications and platforms.

These future improvements will enhance the scalability, usability, and overall performance of the fake news detection system while supporting wider real-world adoption.

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